

VALLEY OAK CUSTOMS

Design Manager

User Manual

Version 1.6.0

October 2026

Find it. Organize it. Open it. Make it.

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1 Product overview and safety model

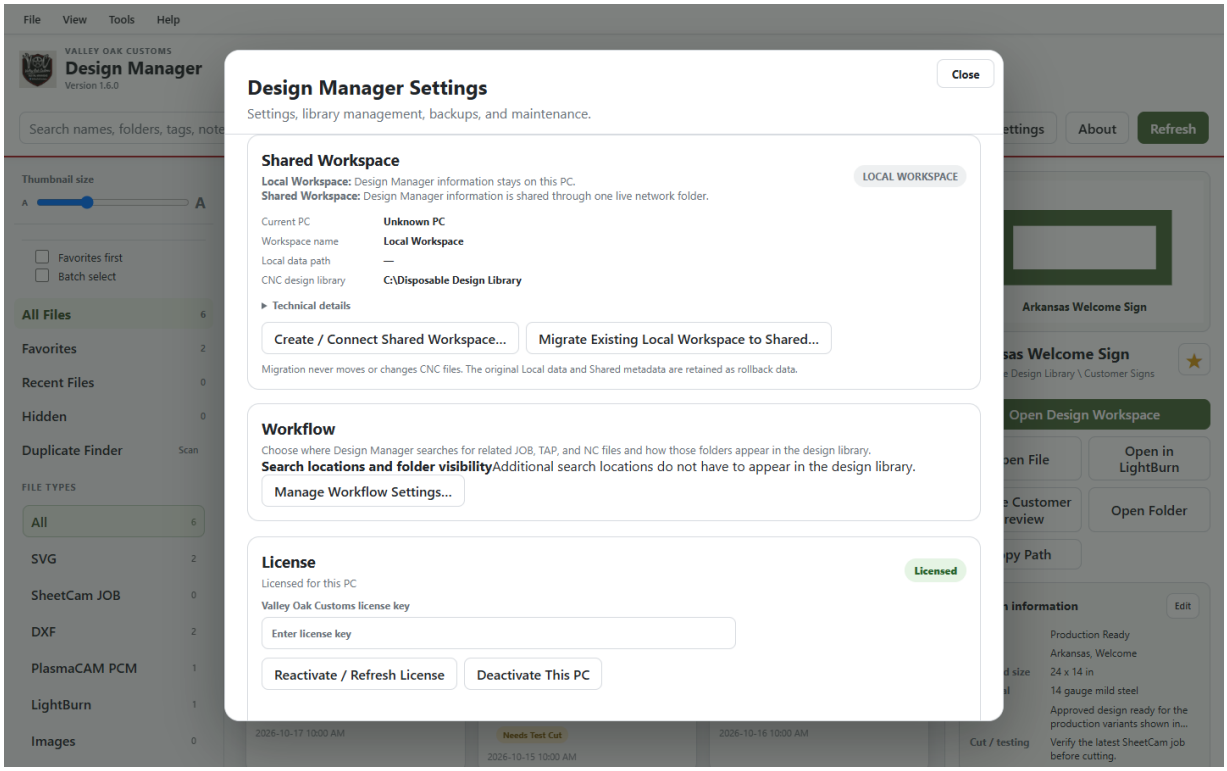
Design Manager catalogs CNC files in their existing Windows folders. It is an organizer and visual browser, not CAD, CAM, or a machine controller.

- Normal browsing, organization, previewing, and migration do not move or change CNC source files.
- TAP and NC files can be inspected or shown in File Explorer but are never executed.
- Rename, Archive, and Move to Recycle Bin are deliberate real-file actions that require confirmation.

2 Installation upgrades and licensing

Design Manager v1.6 requires 64-bit Windows 10 or 11. The installer carries the required .NET 8 runtime; Microsoft Edge WebView2 Runtime remains an external Windows prerequisite.

- Close Design Manager before running Setup. When existing public profile or integration data is present, Setup creates and SHA-256 verifies a timestamped backup under Documents\Valley Oak Design Manager\Pre-Upgrade Backups before replacing application files.
- Backup failure stops Setup before file replacement. Setup does not automatically restore or downgrade a database. Keep the previous installer, independent backups, and the automatic backup until v1.6 is accepted.
- A Local Workspace stays Local. Shared migration begins only when explicitly selected. The installer backup preserves Shared connection configuration but does not traverse or copy Shared network data.
- Use Settings > License to activate or deactivate this PC. The $99 perpetual license covers Design Manager 1.x on up to two PCs and continues working offline after activation. New installations include a 30-day full-feature trial; Viewer Mode preserves safe access afterward. A dead-PC reset is reviewed by Valley Oak support.

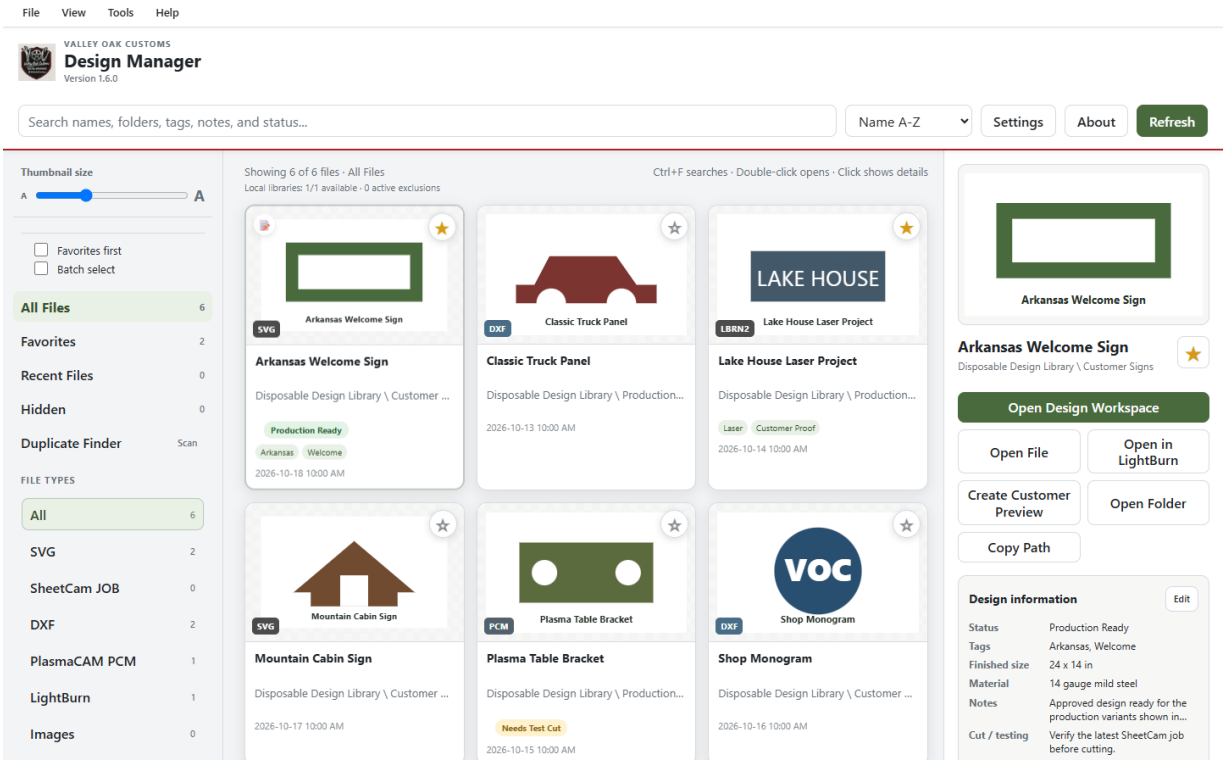


Settings shows licensing, appearance, and Design Library Folder controls.

3 Main screen menus and daily workflow

Use the left filters, center library, and right Details panel together. The Windows-style application menu and the search/sort toolbar remain available while the library scrolls.

- The left sidebar, center gallery, and right details panel have independent scroll regions, including at reduced window heights. Mouse-wheel and scrollbar-drag actions stay within the intended region.
- Search filenames, folders, tags, notes, and status. Select once for details; double-click a supported file to open it.
- The display-first Details panel keeps common actions near the top. Choose Edit when you need to change saved design information.
- For an SVG or DXF, Open Design Workspace provides a design-centered manufacturing view. Its secondary header stays visually joined to the main header while workspace content scrolls beneath it. Back to Library restores the library view.
- Light and dark appearance modes preserve the same commands and scrolling behavior. Menus, popovers, dialogs, and the workflow-refresh reminder remain reachable.



The v1.6 library keeps filters, design cards, and Details independently available.

4 Libraries folders and file types

Design Library Folders supply the normal browser. Workflow locations can contain JOB, TAP, NC, or LBRN2 files without becoming design-card libraries.

- Library cards support SVG, DXF, SheetCam JOB, PlasmaCAM PCM, LightBurn LBRN2, PNG, JPG/JPEG, and WebP. SVG, DXF, images, supported PCM, and LBRN2 projects with embedded thumbnails can display previews.
- PCM support is read-only and limited to validated PCM 1.01 line geometry. It does not edit, repair, convert, or prove machine readiness. Unsupported or malformed PCM files safely fall back without a generated preview.
- LBRN2 support extracts a validated embedded PNG thumbnail only. Missing, invalid, oversized, or unsupported thumbnails fall back safely; Design Manager does not render native LightBurn vectors.
- TAP and NC are G-code Programs and use the dedicated safe machine-file preview. Add Local roots and Include subfolders under Settings > Design Library Folders. Shared roots are Administrator-managed and mapped per PC.

5 Search favorites Controlled Tags collections and status

These tools organize files without moving them.

- Search checks filenames, folders, tags, notes, and status. Filters narrow the visible library.
- Favorites mark frequent work; collections make virtual groups; status records shop readiness.
- Controlled Tags keep suggested names consistent while preserving deliberate user-created tags.

6 File actions and batch tools

Batch organization changes saved metadata. Real-file actions require careful confirmation.

- Hide removes a file from normal browsing.
- Rename changes the disk filename. Archive moves the file. Move to Recycle Bin uses Windows Recycle Bin.
- Shared mode disables unsupported source-file operations rather than risking inconsistent metadata.

7 Design details and shop information

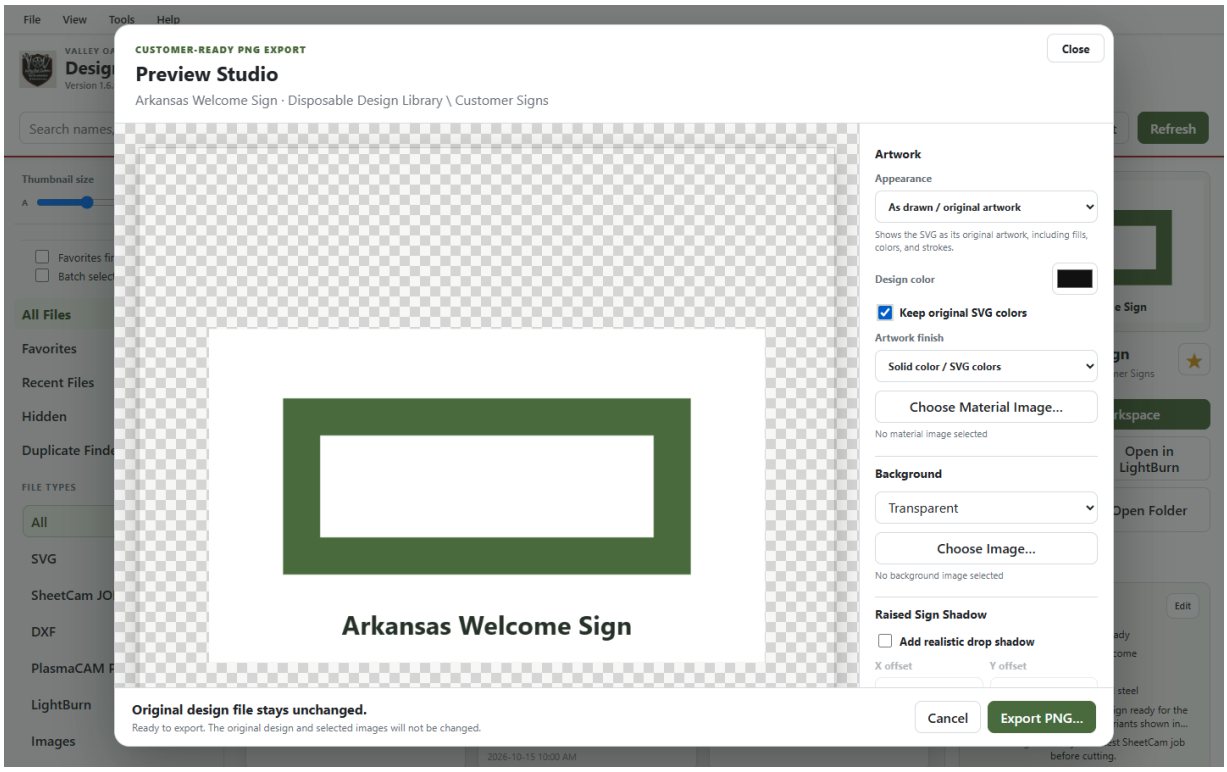
Use the Details panel to record information useful on the next run.

- Store dimensions, material, finish, cut time, test results, and customer notes where useful.
- Saved automatically confirms pending organization changes were written.

8 Preview Studio

Preview Studio exports a separate PNG proof from SVG or DXF.

- Choose artwork, background, branding, proof text, size, and padding.
- For complex DXF contours, As drawn / cut lines is the safest view.
- The source SVG or DXF remains unchanged.

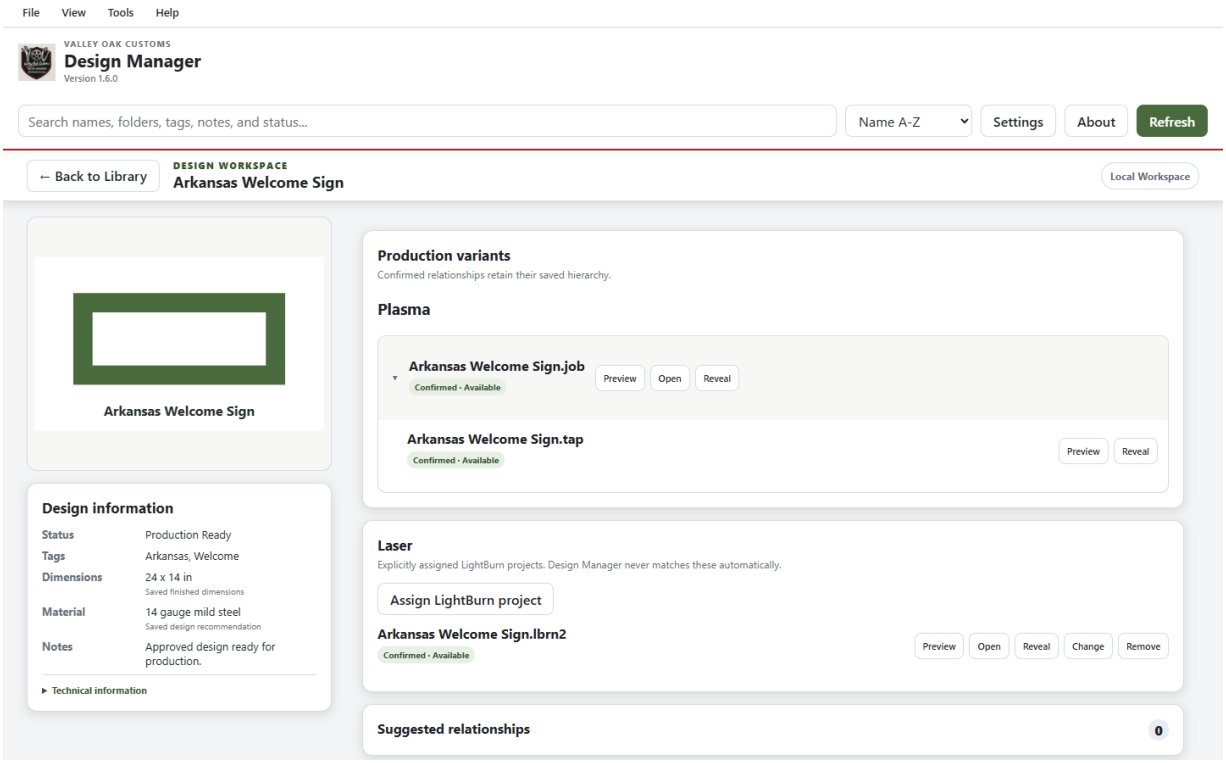


Preview Studio creates a separate proof while the original design stays unchanged.

9 Design Workspace Plasma and Laser

Design Workspace keeps one SVG or DXF anchored while showing saved design information and production variants together.

- Open it from the selected design using Open Design Workspace. Back to Library returns without changing saved relationships.
- The Plasma section keeps confirmed Design to JOB to TAP/NC hierarchy. Direct Design to machine-file links appear separately.
- The Laser section shows one or more LBRN2 projects assigned explicitly to the design. Use Assign LightBurn project, Change, or Remove. Filenames are never used to confirm LightBurn project relationships automatically.
- Suggested Plasma relationships remain visually separate and require Review before they become active. Preview and Reveal use the existing safe file actions. Returning from Preview, minimizing, or switching windows preserves the current view.

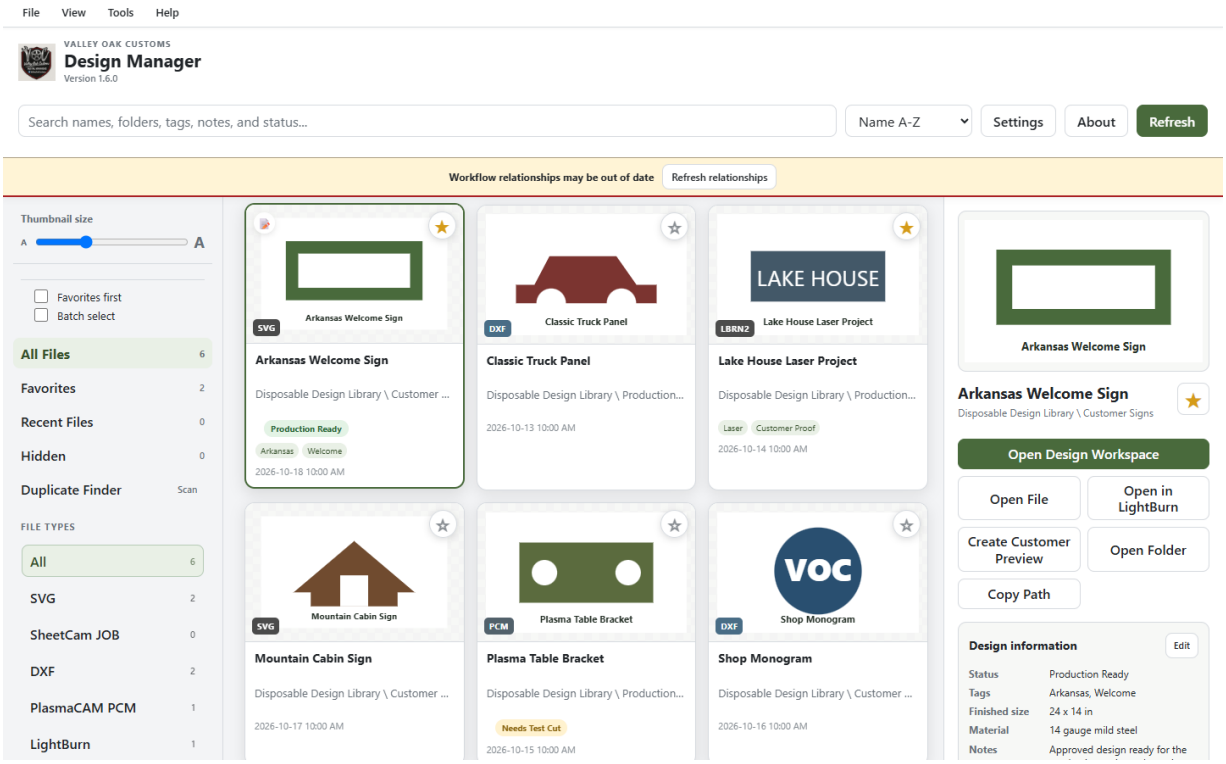


Design Workspace shows confirmed Plasma files and explicitly assigned LightBurn projects.

10 Related Files discovery and refresh

Related Files preserves the many-to-many reality of shop work. Discovery creates review candidates; it does not make inferred relationships authoritative.

- A JOB can contain several Designs. A Design can belong to several JOBs. A JOB can have several TAP or NC programs. An LBRN2 relationship is always an explicit user choice.
- Library Refresh updates the main file index. It does not automatically start a second workflow scan or confirm relationships. When file changes make relationship data potentially stale, the visible reminder offers Refresh relationships as the explicit recheck.
- Design Manager uses filenames, locations, file types, and bounded context to suggest Plasma candidates. Manual decisions outrank later discovery.
- When a confirmed Design to JOB connection already explains a machine program, redundant direct Design to machine suggestions are kept out of normal review.



The reminder offers an explicit relationship refresh after the library changes.

Relationship history: Design Manager keeps earlier choices so you can restore a relationship later.

11 Needs Review

Needs Review contains suggestions that still require a decision.

- Open it from a selected design to see direct suggestions plus unresolved machine-file suggestions beneath that design's confirmed JOBs. Choose Show all needs review for the global queue.
- Sort columns and inspect both paths, types, dates, context, and match reasons.
- Choose Confirm to make the pair active or Not related to retain an explicit rejection. A reconsidered pair remains inactive until Confirm succeeds.

12 All Workflow Files

All Workflow Files includes workflow-only locations and files absent from the normal browser.

- Launch from a selected design to keep its confirmed manufacturing context. Choose Show all workflow files for the global list.
- Use Find and sortable columns in large libraries.
- Preview or Show in folder safely before deciding.

13 Saved Relationships and states

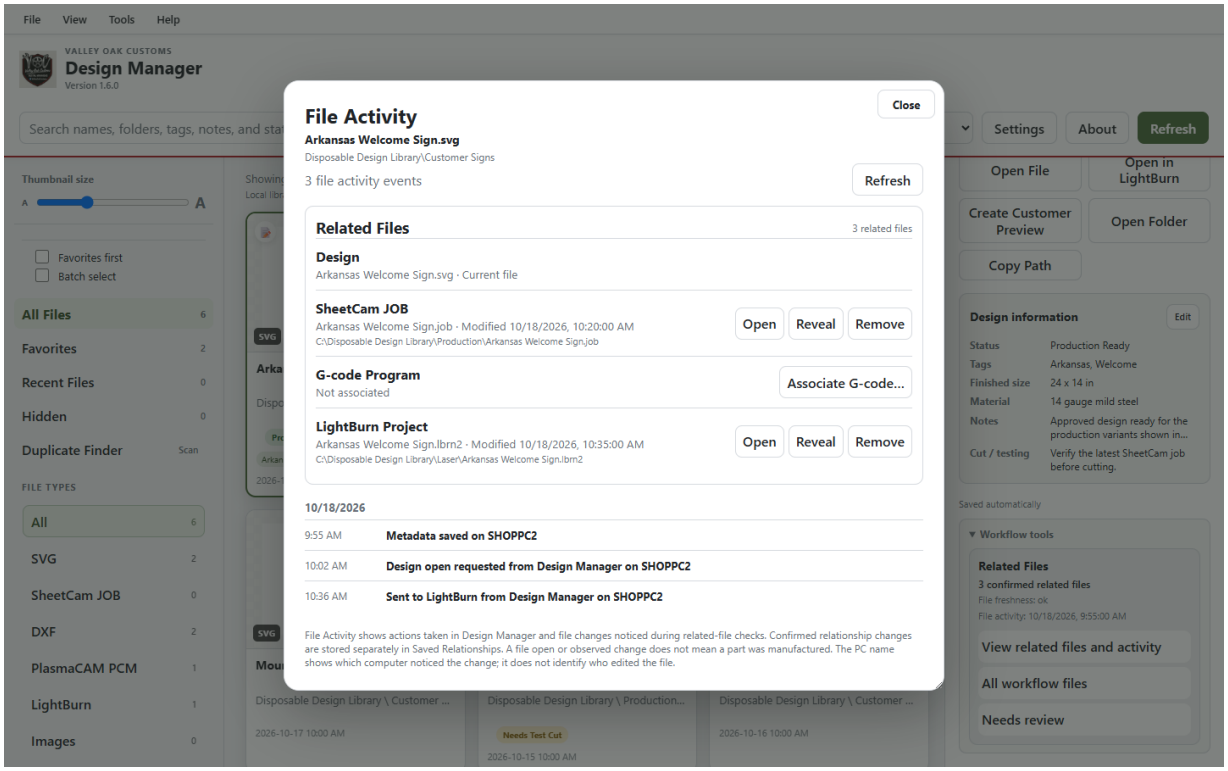
Saved Relationships retains decisions and history for each pair.

- Confirmed is active. Removed is retained history and is inactive.
- Reconsidered returns an existing pair to Needs Review; it remains inactive until confirmed.
- Superseded preserves an older choice replaced by a correction. Restore relationship reactivates a saved connection when allowed.
- Choose different file uses the protected correction flow so old and replacement states remain traceable.

14 File Activity and relationship history

View related files shows the selected path, current-file role, active related items, and recorded File Activity.

- A JOB shows its directly related Designs and G-code. A Design shows its direct JOBs, appropriate G-code through them, and explicitly assigned LightBurn projects.
- Use Add or Associate actions to connect a file manually when automatic matching is not enough. LightBurn project assignment always uses this explicit path.
- File Activity records actions taken in Design Manager and file changes noticed during related-file checks. A file open or observed change does not mean a part was manufactured. The PC name shows which computer noticed the change, not who edited the source file.
- Confirmed relationship changes are stored separately in Saved Relationships. History remains available when a relationship is removed, reconsidered, or superseded.



Related Files and File Activity show saved context without claiming a part was manufactured.

15 Choose a related file

The picker keeps the current file fixed while you choose the other side.

- Likely related opens first and ranks compatible candidates using existing match evidence.
- Preview inspects a candidate through its safe existing preview path. Show in folder reveals the exact selected file.
- Show all compatible files restores the complete searchable list for unusual or historical filenames.
- Files with the same name under different roots remain distinct. Repeat Add another Design for mixed-sheet JOBs.

16 TAP and NC safe previews

The machine-file preview renders supported TAP and NC geometry without executing machine instructions.

- Use fit, pan, and the traverse display to inspect the visual path.
- A preview is not CAM verification, postprocessor approval, machine simulation, or proof that a program is safe to run.
- Design Manager does not provide an Open or Run action for TAP or NC programs.

17 Library Refresh relationship refresh and search

Normal browsing and related-file search are separate choices.

- Local folder exclusions hide selected subfolders from normal scans without deleting files or saved organization.
- Library Refresh updates the library cards, counts, previews, and file index. Refresh relationships explicitly rechecks configured workflow locations and saved discovery state.
- A workflow location can be searched for related files without appearing as a normal design library. After changing Local search locations, use Refresh relationships to apply the settings.
- The workflow-refresh reminder remains visible when a library change may have made relationship data stale. Dismiss it only when you deliberately do not need a relationship scan.
- Local exclusions are unavailable while Shared Workspace is active; Shared roots are Administrator-managed.

18 Local and Shared behavior

The same organization and relationship terms apply in both modes.

- Local choices stay in the current PC profile. A normal upgrade does not migrate them.
- Shared metadata and Workflow decisions converge between registered PCs with conflict protection.
- Unavailable or incompatible Shared storage preserves the view and blocks unsafe writes.

19 Shared Workspace setup and Local migration

Settings offers Create / Connect Shared Workspace and Migrate Existing Local Workspace to Shared.

- Migration first shows a review preview, preserves the source fingerprint, and creates a complete Local backup before staged Shared cutover. Canceling before approval leaves Local authoritative and the destination untouched.
- Migration imports supported Local organization and Workflow state without moving CNC files. Create Empty Shared Workspace does not copy Local organization.
- Connect joins an existing workspace after connection and compatibility checks.

20 Multiple Shared library roots

A Shared Workspace can have more than one authoritative library root.

- The Administrator adds, removes, and orders authoritative roots. Root identity and ordering are shared.
- Each PC maps every Shared root to the local drive or UNC path that reaches the same content. Paths may differ; root meaning must match.
- During migration, review each source-to-destination root mapping. Unmapped or ambiguous roots stop safely.

21 Administrator Members and recovery

The first authorized Shared device becomes Workspace Administrator; other registered PCs are Members.

- Members perform ordinary Shared organization and Workflow work. The Administrator manages roots, Shared settings, upgrades, backups, devices, recovery designation, and authority transfer.
- Designate at least one Member as a Recovery Device. If the Administrator PC is lost, an eligible Recovery Device can use the protected Recover Administrator flow.
- Stale administrator or device changes are rejected and require Refresh instead of overwriting newer work.

22 Shared offline conflicts and startup

Shared mode reuses verified saved discovery state at startup and reconciles library changes separately.

- Opening Workflow does not require a full rescan when valid state exists. Refresh related files remains the explicit recheck action.
- If Shared metadata storage is unavailable, Design Manager changes to OFFLINE - READ ONLY. Shared edits are blocked and are not queued.
- After access returns, Refresh reads current state. Stale writes are rejected instead of losing another PC’s update.
- A slow or unavailable CNC library is reported separately when Shared metadata itself remains reachable.

23 Shared upgrades returning and conversion

The Administrator can upgrade an eligible Shared Workspace after a verified backup.

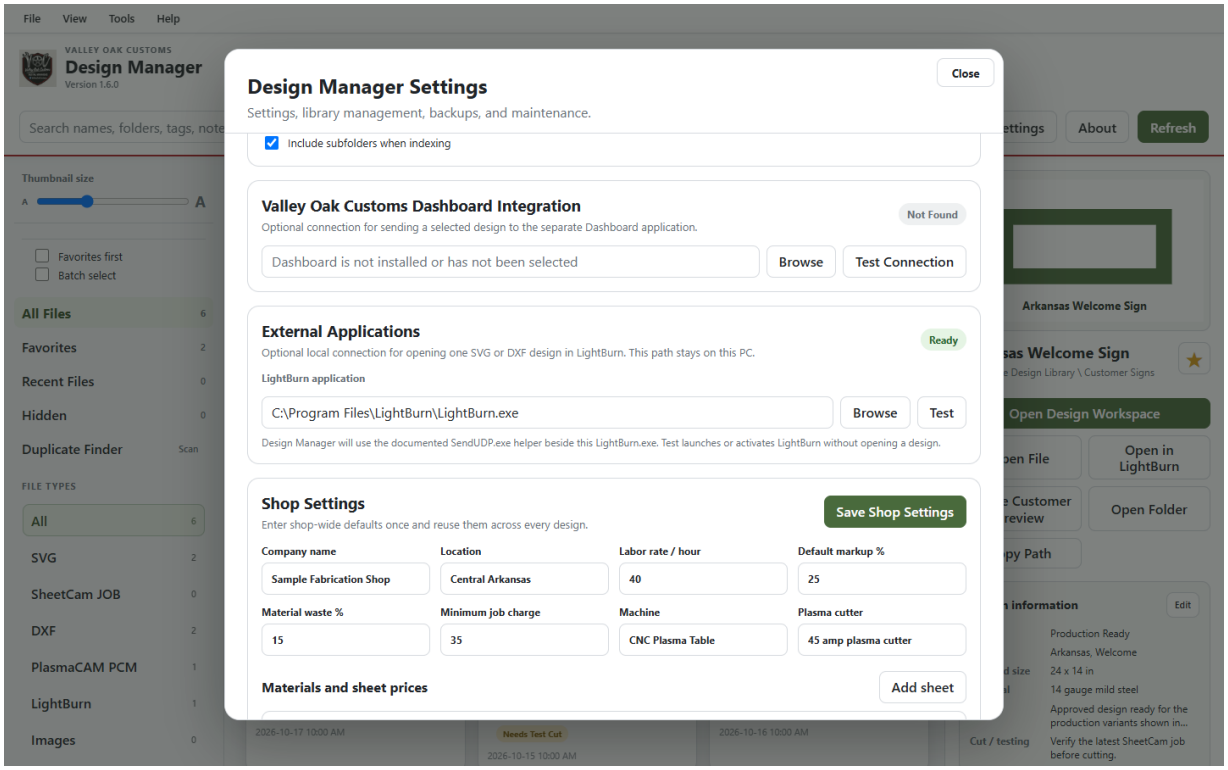
- Update participating PCs before enabling newer Shared Workflow features. Older incompatible clients fail closed or remain read-only.
- Return to Previous Local Workspace creates a verified Shared backup, reconnects this PC to its preserved earlier Local profile, and does not import later Shared changes.
- Convert Shared Workspace to New Local creates a separate Local copy from current Shared metadata after backup; it does not overwrite the previous Local Workspace.

Plan the upgrade: Update participating PCs before normal Shared use resumes after a workspace upgrade.

24 Settings LightBurn Dashboard and updates

Settings contains libraries, Workflow, appearance, licensing, integrations, updates, and maintenance.

- Configure and Test Connection for LightBurn per PC. Open in LightBurn supports SVG and DXF only and leaves the source unchanged. LBRN2 projects use their normal Windows file association when opened.
- Dashboard integration is configured per PC when the supported Valley Oak dashboard is installed.
- Check for Updates is manual. It sends a read-only request only when selected and opens the official update page only after a verified result.



Per-PC Dashboard and LightBurn integration settings.

25 Automatic pre-upgrade backup and recovery

The public v1.6 installer requires a complete verified backup before it replaces files on an existing installation.

- The timestamped package is stored outside the installation directory under %USERPROFILE% \Documents\Valley Oak Design Manager\Pre-Upgrade Backups. LAST-PRE-UPGRADE-BACKUP.txt records the newest path and manifest hash.
- The backup includes the complete production profile, local integration state, and a configured external Local Workspace data directory. It includes licensing state as protected files without printing license contents.

- Shared Workspace connection configuration in the local profile is preserved, but Shared network workspace data is never traversed, copied, modified, migrated, or restored by the installer backup.
- To recover, first verify the package with its included Windows PowerShell 5.1 verification script. Reinstall the exact previous version without launching it, preserve the current state by renaming it, and restore each manifest target manually. Never open a v1.6-migrated database with an unsupported v1.5 client.

26 Backups export restore and storage locations

Design Manager backups protect settings, Workflow choices, and organization; source CNC files need their own independent backup.

- Local export and restore operate on Local profile data. Local state is under %LOCALAPPDATA%\Valley Oak Customs\Design Manager; local integration state is under %LOCALAPPDATA%\Valley Oak Customs\Integration.
- Shared backups are manual and metadata-only. Shared restore validates a complete backup and writes to a new or empty recovery directory; it never overwrites the active workspace.
- Shared metadata is in the selected .valleyoak-workspace folder. Per-PC mappings and integration paths remain local. Use Open Backups or Open Shared Backups to locate recovery packages.

27 Repair diagnostics and troubleshooting

Repair previews stale references and protects unavailable folders. Diagnostics gathers support information without copying design files.

- Start with Library Refresh, clear Search and filters, verify Windows access, and read the visible status message. Use Refresh relationships when related-file data is stale.
- Reconnect drives and shares before Repair; cancel when an expected folder is unavailable. Shared offline: restore access, then Refresh. Do not create a replacement workspace over an unavailable share.
- Missing related file: inspect Needs Review, All Workflow Files, and Saved Relationships. Missing PCM or LBRN2 artwork can mean the source is unsupported or the LBRN2 project lacks a usable embedded thumbnail.
- Support: include the version from Help > About, steps, exact message text, PC name, and reviewed diagnostics.

Appendix A Status and relationship reference

Use consistent values so history stays understandable.

Customer term	Meaning
Current	The current file generation.
Older file	A historical file generation.
Confirmed	An active saved relationship.
Removed	Inactive retained history.
Reconsidered	Inactive and returned to Needs Review.
Superseded	An older choice replaced by a newer correction.

Appendix B Backup checklist

Verify every layer before a PC move, upgrade, or recovery.

- Actual CNC folders have an independent backup.
- The Design Manager backup or installer pre-upgrade package verifies successfully.
- The previous installer and the current backup are retained until acceptance.
- Expected drives and shares are online before Repair. Shared network data has its own backup plan.

Appendix C Glossary

The main Workflow uses ordinary shop terms.

Term	Plain-language meaning
Design	SVG or DXF artwork.
SheetCam JOB	A saved SheetCam project.
G-code Program	TAP or NC machine output.
PCM preview	Read-only PCM 1.01 line geometry.
LightBurn project	An explicitly assigned LBRN2 file.
Related Files	Saved connections between files.
Design Workspace	A design-centered view of saved information and Plasma/Laser variants.